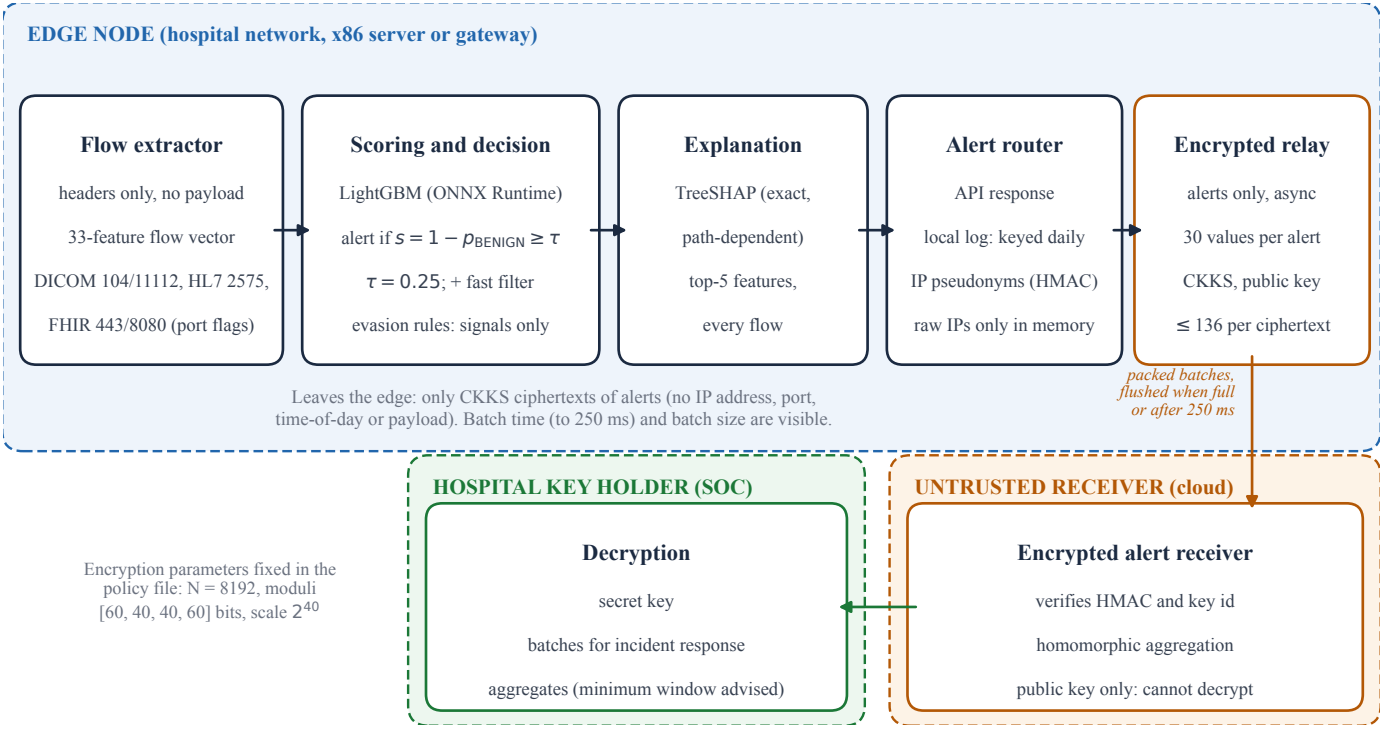

EDGE NODE (hospital network, x86 server or gateway)
Flow extractor
headers only, no payload
33-feature flow vector
DICOM 104/11112, HL7 2575,
FHIR 443/8080 (port flags)
Scoring and decision
LightGBM (ONNX Runtime)
alert if $s = 1 - p_{\text{BENIGN}} \geq \tau$
$\tau = 0.25$; + fast filter
evasion rules: signals only
Explanation
TreeSHAP (exact,
path-dependent)
top-5 features,
every flow
Alert router
API response
local log: keyed daily
IP pseudonyms (HMAC)
raw IPs only in memory
Encrypted relay
alerts only, async
30 values per alert
CKKS, public key
≤ 136 per ciphertext
packed batches,
flushed when full
or after 250 ms
Leaves the edge: only CKKS ciphertexts of alerts (no IP address, port,
time-of-day or payload). Batch time (to 250 ms) and batch size are visible.
HOSPITAL KEY HOLDER (SOC)
Decryption
secret key
batches for incident response
aggregates (minimum window advised)
UNTRUSTED RECEIVER (cloud)
Encrypted alert receiver
verifies HMAC and key id
homomorphic aggregation
public key only: cannot decrypt
Encryption parameters fixed in the
policy file: N = 8192, moduli
[60, 40, 40, 60] bits, scale 2^{40}